

# Kaustub P Anand

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## RESEARCHER BIO

I am a PhD candidate at Purdue University in the Department of Physics and Astronomy and work in the Department of Earth, Atmospheric, and Planetary Sciences. My research interests are in solar system dynamics, especially planetary rings and moons, comets, and small bodies (ex: Centaurs). I primarily work with numerical simulations and have done observations in the past.

## TECHNICAL SKILLS

**Coding** : Python, FORTRAN, NetCDF, HDF5, SHTools, Java, IDL,  $\text{\LaTeX}$ , SAOImage DS9, C, C++, Matlab, IRAF

**Physical Effects Coded**: Gravitational harmonics, collisional fragmentation, radiation forces including Yarkovsky and the Yarkovsky-Schach effect, 1D fluid debris rings, hybrid fluid ring – N body code

## EDUCATION

**Purdue University** Aug 2021 – Present  
*Doctor of Philosophy (PhD) in Physics.*  
Thesis: Understanding the origins and dynamics of rings and corresponding satellites with application to Centaurs and the Martian moons.  
Advisor: Dr. David A. Minton

**Purdue University** Aug 2017 – May 2021  
*Bachelor of Science in Physics Honors.*  
Minors in Astronomy and Math; Graduated with Distinction.  
Cum. GPA: 3.92/4.00; Major GPA: 3.96/4.00

## PUBLISHED RESEARCH PAPERS

**Ejected Surface Regolith as a Potential Source Material for Centaur Rings** doi:10.3847/PSJ/ae7724  
*Kaustub P. Anand, David A. Minton, Julie Brisset*

**The Sesquinary Catastrophe on Deimos Can Reconcile Its Excited Past with Its Dynamically Cool Present** doi:10.3847/PSJ/ae2fba  
*Kaustub P. Anand, Matija Ćuk, David A. Minton*

**Two Possible Orbital Histories of Phobos** doi:10.3847/PSJ/adc1ba  
*Matija Ćuk, Kaustub P. Anand, David A. Minton*

**Galaxy populations in protoclusters at cosmic noon** doi:10.1051/0004-6361/202452628  
*Moirá Andrews, M. Celeste Artale, Ankit Kumar, Kyoung-Soo Lee, Tess Florek, Kaustub P. Anand, et al.*

**Swiftest: An N-body Integrator for Gravitational Systems** doi:10.21105/joss.05409  
*Carlisle Wishard, Jennifer Pouplin, Jacob Elliott, Dana Singh, Kaustub P. Anand, David A. Minton*

## OTHER RELEVANT RESEARCH EXPERIENCE

**Evolving a fluid Martian ring under the Yarkovsky-Schach effect** Feb 2026 – Present  
*Advisors: Dr. David A. Minton and Dr. Matija Ćuk*  
*Purdue University*

**Star formation in dust-obscured Galaxy Clusters in far-IR & protoclusters in simulation** May 2019 – May 2022  
*Advisors: Dr. Kyoung-Soo Lee and Dr. Maria Celeste Artale*  
*Purdue University*

## CONFERENCES AND INVITED TALKS

**Colloquium Seminar at Southwest Research Institute (SwRI)** Nov 2026

**AAS Division on Dynamical Astronomy (DDA)** 2024 - 2026

**AAS Division of Planetary Sciences (DPS)** 2024, 2026

**Purdue EAPS Planetary Science Seminar (1 talk/year)** 2023 - 2026

**Gordon Research Seminar & Conference: Origins of the Solar System** June 2025

**NOIRLab: The Solar System in Context** Sept - Oct 2025

**Division of Planetary Sciences-Europlanet Science Congress** 2023, 2025

**Astronomy Seminar at the University of Connecticut** Feb 2025

**Outreach Talk at Nahar International School, Mumbai, India** Dec 2025

**Purdue University Department of Physics and Astronomy, and EAPS 1-minute colloquium**

2022 - 2026

**Purdue University Department of Physics and Astronomy annual poster symposium**

2023

**Purdue Fall Research Expo**

Fall 2020

## LEADERSHIP AND PROFESSIONAL EXPERIENCE

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**Local Organizing Committee Member for DDA 2027**

Sept 2026 - Present

*LOC member*

Serving as a member of the Local Organizing Committee for the Division on Dynamical Astronomy (DDA) 2027 meeting in Bloomington, IN

**JPL Planetary Science Summer School**

May - July 2026

*Proposal Manager (Management Hex Role)*

*Jet Propulsion Laboratory*

Designed a New-Frontiers class NASA mission on Ceres called Proserpina in 11 weeks as a part of the JPL Mission Design Schools. I helped with personnel management, moderated conflict resolutions, managed graphics deliverables, and maintained flexibility to help the PI resolve last-minute problems.

**Graduate Student Ombudsperson**

May 2026 - Present

*Ombudsperson*

*Purdue University*

Served as the graduate student ombudsperson for the Department of Earth, Atmospheric, and Planetary Science to listen and help student grievances and conflicts.

**NASA Review Panel Executive Secretary**

2026

*Executive Secretary*

*Online*

Served as the executive secretary on 2 NASA proposal review panels for the NASA Science Mission Directorate.

**EAPS Planetary Science Seminar Organizer**

May 2025 - 2026

*Crater Café organizer*

*Purdue University*

Managed and organized the schedule and speakers of Crater Café, the weekly Purdue EAPS planetary science seminar.

**Physics & Astronomy Teaching Assistant**

Aug 2018 - Mar 2020 & Aug 2021 - Dec 2022

*Teaching Assistant*

*Purdue University*

Led and aided students in labs and recitations for modern mechanics (PHYS 172), electricity and magnetism (PHYS 272), senior level Computational Physics Lab (PHYS 580), and Optics Lab (PHYS 450). Started as a Graduate Teaching Assistant in Aug 2021.

**Saturday Morning Astrophysics (SMAP)/Physics Inside Out (PIO)**

Jan - Mar 2020; Jan 2021 - May 2022; June 2025 - 2026

*Outreach Teacher*

*Purdue University*

Teach and introduce common astronomy and physics concepts to elementary & middle school students through hands-on experiments monthly on Saturdays (SMAP) and in the June annually (PIO).

**Peer Success Coach**

Jan 2021 - May 2021

*Tutor*

*Purdue University*

Tutor and coach students at the Purdue Academic Success Center to improve their studying behaviour and motivate them to work towards achieving their current and new academic goals.

**International Student Ambassador**

May 2020 - May 2021

*Ambassador*

*Purdue University*

Aid the Purdue International Students and Scholars (ISS) office as a student ambassador in organizing events for the international student community on campus including new student orientations, cultural mixers, Q&A sessions, etc.

**Global Science Partners Leadership**

Aug 2019 - May 2021

*Team Leader*

*Purdue University*

Organized and led events to analyze the effects of inter-cultural interactions in professional and scientific collaborations among the members of Global Science Partners learning community (~160 students)

## AWARDS & CERTIFICATES

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- Purdue Graduate Student Government Travel Grant (2025)
- Purdue EAPS Departmental Travel Grant (2025)
- Hartmann Student Travel Grant for DPS (2024)
- Martin C. Jishcke Outstanding International Student Award (2021)
- Learning Beyond the Classroom Certificate for intercultural learning in Science (2021)
- Dean's List and Semester Honors (2017 - 2021)
- Honors College (2018 - 2021)
- Arthur N. Pozner Memorial Scholarship for academic excellence in Physics (2020)